

Attention-aware task scheduling under partial observability of human cognitive state: a memory-augmented deep reinforcement learning approach

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Abstract

TBD

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1. Introduction

Scheduling decides which piece of work is executed when, and it is one of the oldest problems in operations research and computer engineering [1–4]. Classical dispatching rules such as First-In-First-Out (FIFO), Shortest Processing Time (SPT) and Earliest Deadline First (EDF) are attractive because they are transparent, cheap to compute and, under the right assumptions, provably optimal: EDF minimises maximum lateness on a single resource with constant capacity [2, 5], and SPT minimises mean flow time [1]. The common assumption underlying these guarantees is that the resource executing the work has a *fixed, state-independent processing capacity*. This assumption is reasonable for processors and machines but is violated whenever the resource is a person.

Human work capacity is dynamic. Sustained effort depletes attentional resources and produces the vigilance decrement first documented by Mackworth [6]

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21 and since replicated across many tasks [7–9]; mental fatigue accumulates with
22 time-on-task and degrades both speed and accuracy [10–13]; and alertness follows
23 a circadian rhythm with a pronounced early-afternoon dip [14–17]. Rest breaks
24 partially reverse these effects, and field studies consistently report that well-timed
25 breaks preserve or even increase productivity despite the time they consume [18–
26 22]. A schedule that is optimal for a constant-capacity machine can therefore be
27 poor for a human: it front-loads demanding work regardless of cognitive state,
28 never allocates recovery, and drives the worker into a low-capacity regime in
29 which deadlines are missed precisely because capacity was ignored. The concern
30 is not only productivity. Chronic fatigue is associated with errors and accidents
31 [23, 24], and the move from Industry 4.0 towards a human-centric Industry 5.0
32 explicitly calls for production and service systems that adapt to the well-being of
33 their workers rather than the reverse [25–29].

34 Operations-management research has long recognised this gap. Models of
35 worker fatigue and recovery have been integrated into dual-resource-constrained
36 systems [30–32], rest-allowance and break-scheduling problems have been formu-
37 lated analytically [33–35], and taxonomies for merging scheduling theory with
38 human factors have been proposed [36–39]. These approaches, however, usually
39 assume that the worker’s state is known or deterministic and optimise an offline
40 plan; they do not provide an adaptive *policy* that reacts to the evolving, uncertain
41 state of a specific person during the day. Deep reinforcement learning (DRL),
42 in turn, has produced effective adaptive schedulers for job shops, clusters and
43 production lines [40–46], and recent surveys document rapid growth of the field
44 [47–49]. Yet the resources in these studies are machines, and the few recent studies
45 that include worker fatigue embed it in large shop-floor models where fatigue is a

deterministic, fully observed variable [50, 51]. Two features of the human problem are thus rarely addressed together: *the decision of when to rest is itself a scheduling action*, and *the cognitive state that should drive that decision cannot be measured directly*. In practice, attention and fatigue are inferred from noisy proxies such as physiological signals, eye tracking, performance probes or self-report [52–56], which makes the scheduling problem partially observable.

This paper studies attention-aware task scheduling for a single knowledge or production worker as a partially observable Markov decision process (POMDP) [57–59] and solves it with a memory-augmented deep Q-network. We pay particular attention to the methodological issues that make DRL comparisons hard to trust [60, 61]: we compare against strong, *tuned*, cognition-aware heuristics rather than only against FIFO and EDF; we use disjoint seed sets for training, tuning and testing; we evaluate on 500 held-out workdays with common random numbers; we report ten independent training seeds with confidence intervals and multiplicity-corrected paired tests; and we separate effects of the reward function from effects of the dynamics through ablation.

The contributions are as follows.

- (i) **A cognitively grounded scheduling simulator.** We formulate a workday-scale scheduling environment in which a latent homeostatic attention resource, an exponentially accumulating and recovering fatigue state [30], and a circadian process [14, 15] jointly determine the worker’s instantaneous work rate. Tasks arrive stochastically with heterogeneous difficulty, work content and deadlines. Rest is an explicit action. The simulator, its parameters and their behavioural rationale are fully specified and released.
- (ii) **A POMDP formulation with a multi-objective reward.** The scheduler ob-

71 serves only noisy estimates of attention and fatigue together with a compact,
72 size-invariant summary of the task queue. The reward combines deadline-
73 weighted task value with explicit attention and fatigue terms, and we analyse
74 the resulting productivity–fatigue trade-off as a Pareto front [62, 63].

75 (iii) **An attention-aware DQN (AA-DQN).** We combine a finite observation
76 history, which acts as an approximate sufficient statistic for the belief state
77 [64–66], with Double Q-learning [67] and a dueling architecture [68]. A
78 component ablation quantifies the contribution of each element.

79 (iv) **A rigorous empirical study.** Against seven baselines, including grid-tuned
80 threshold and attention-matching heuristics that observe the same noisy
81 cognitive signals, we test (a) whether learning helps, (b) *why* it helps, through
82 environment ablations that remove observation noise, the circadian process,
83 the cognitive dynamics or the cognitive reward terms, (c) what behaviour
84 is learned, and (d) how the learned policy generalises to unseen worker
85 profiles, workloads and sensor-noise levels.

86 The remainder of the paper is organised as follows. Section 2 reviews related
87 work and positions the study. Section 3 formulates the problem. Section 4 presents
88 AA-DQN. Section 5 details the experimental protocol. Section 6 reports results,
89 Section 7 discusses implications, ethics and limitations, and Section 8 concludes.

90 **2. Related work**

91 *2.1. Classical and dynamic scheduling*

92 Single-resource sequencing theory provides the canonical dispatching rules
93 and their optimality conditions: SPT for mean flow time [1], EDF for maximum

lateness [2], and the Moore–Hodgson algorithm for the number of late jobs [69]. The three-field classification of Graham et al. [3] and the textbook treatment of Pinedo [4] organise the deterministic theory. In real-time computing, EDF is optimal for preemptive scheduling of independent tasks whenever the processor is not overloaded [5], but its performance can collapse under overload, when a domino effect causes many deadlines to be missed [70]. This failure mode is directly relevant here, because a fatigued worker is in effect an overloaded resource. Surveys of priority rules [71, 72] and of dynamic and reactive scheduling [73, 74] show that practice relies heavily on such rules when information arrives online. In all of these models, however, processing times are exogenous: they do not depend on the history of the schedule.

2.2. *Human factors in scheduling and workforce models*

Personnel scheduling and rostering [75–77] assign workers to shifts but typically treat performance within a shift as constant. Research at the interface of operations and human resources argues that this abstraction misses first-order effects [37, 38], and Lodree et al. [36] proposed a taxonomy for integrating human factors into scheduling theory. Quantitative models of learning, forgetting, fatigue and recovery have been developed for dual-resource-constrained systems [30–32, 78]. They use exponential fatigue accumulation during work and exponential recovery during rest, which is the functional form we adopt. Rest-break scheduling has been formulated as an optimisation problem since Bechtold et al. [33], with ergonomic guidance on work–rest ratios summarised by Konz [34] and rest-allowance models for task assignment developed by Calzavara et al. [35]. Fatigue has also been incorporated into flexible job-shop formulations solved by metaheuristics [50] and, very recently, by graph-based DRL [51]. Reviews of hu-

man factors in Industry 4.0 and in order picking [39, 79–81] conclude that worker heterogeneity and well-being are still under-represented in operational models. A common feature of this literature is that the worker state is assumed to be known, either deterministic or computed from the schedule, so the decision problem is fully observable.

2.3. *Deep reinforcement learning for scheduling*

Reinforcement learning has been applied to scheduling since the mid-1990s [82, 83], and the combination of Q-learning [84] with deep function approximation [85] enabled scheduling policies that operate on rich state representations. Notable examples include cluster resource management [40], data-processing job scheduling with graph neural networks [41], semiconductor production scheduling [42], dynamic flexible job shops with job insertions [43, 46], and size-agnostic job-shop dispatching [44, 45, 86]. Parameter-adaptive RL for dynamic job shops [87] and RL for combinatorial optimisation more broadly [88, 89] complete the picture. Systematic reviews [47–49] note three recurring weaknesses: comparisons against untuned or weak baselines, limited statistical analysis, and little attention to human resources. Our study addresses all three.

2.4. *Cognitive state, its measurement, and adaptive systems*

Resource theories of attention [90, 91] and cognitive load theory [92] model mental capacity as a limited resource. Vigilance research shows that this resource is depleted by sustained effort [6, 7] and partially restored by brief breaks [93]. Accounts of fatigue as an effort-regulation or opportunity-cost signal [12, 94] and evidence linking fatigue to disengagement [95] motivate treating fatigue as a state variable that shapes future capacity. Biomathematical fatigue models, from

143 the two-process model of sleep regulation [96–98] to the three-process model of
144 alertness [14] and applied models such as SAFTE [99], combine homeostatic and
145 circadian components. We borrow this decomposition at the time scale of a single
146 workday. Recovery research in organisational psychology confirms the value of
147 micro-breaks [22, 100–102]. On the measurement side, workload and fatigue can
148 be estimated from EEG, cardiac, ocular and behavioural signals [52–54, 103, 104],
149 from vigilance probes such as the PVT [55, 105], or from questionnaires such as
150 NASA-TLX [56, 106]. All of these estimates are noisy. Adaptive automation
151 and physiological computing use such estimates to adjust system behaviour [107–
152 112], and interruption-management systems decide when to notify a user based
153 on inferred cognitive state [113–116]. Reinforcement learning has been used
154 to personalise just-in-time health interventions [117–120], and in human–robot
155 collaboration to adapt task allocation to human capability and fatigue [121–125].
156 These works show that cognitive-state estimates can drive decisions, but they rarely
157 address the sequential scheduling of a task queue under deadlines.

158 *2.5. Partial observability and multiple objectives in RL*

159 When the state is not directly observed, the optimal policy is a function of the
160 belief state [57–59]. Memoryless policies can be arbitrarily poor [64]. Practical
161 deep RL approximates the belief either with recurrent networks [65, 126, 127]
162 or with finite observation histories, as in the frame stacking of Mnih et al. [85].
163 Ni et al. [66] show that well-tuned memory-based model-free agents are strong
164 POMDP baselines. Stabilisation techniques for DQN, namely experience replay
165 [128], target networks [85], Double Q-learning [67], dueling heads [68] and their
166 combinations [129], are well established, as is the theory of TD learning with func-
167 tion approximation [130]. Scheduling for humans is inherently multi-objective:

168 productivity, deadlines and well-being conflict. Multi-objective RL provides the
 169 vocabulary of scalarisation and Pareto fronts [62, 63, 131]. Reward shaping must
 170 be used with care, because additional reward terms can change the optimal policy
 171 [132]. Finally, the deployment of RL in settings that involve people raises safety
 172 and reliability concerns [133–135], which we address in Section 7.

173 2.6. *Research gap*

174 Table 1 positions this work against representative streams. To our knowledge,
 175 no prior study simultaneously (i) models dynamic attention, fatigue and circadian
 176 variation as endogenous state, (ii) treats rest as a scheduling action, (iii) assumes
 177 that the cognitive state is only partially observable, (iv) learns an adaptive policy,
 178 and (v) benchmarks that policy against tuned, cognition-aware heuristics with
 179 multi-seed statistical testing. Our study fills this gap.

180 3. Problem formulation

181 We consider a single worker who processes a stream of tasks during one work-
 182 day. A scheduler, which could be a digital assistant, a workflow-management
 183 system or a supervisory controller in a human–robot cell, decides at regular inter-
 184 vals what the worker should do next: start or continue an easy task, a hard task, or
 185 take a short break. The worker’s capacity depends on a latent cognitive state that
 186 evolves with the schedule. Table 2 summarises the notation and Table 3 lists the
 187 parameter values.

188 3.1. *Workload model*

189 The day is divided into $T = 48$ slots of $\Delta = 10$ minutes (09:00–17:00). Each
 190 day contains $N = 12$ tasks. Six tasks are available at the start of the day and

Table 1: Positioning of this study relative to representative research streams (✓: addressed; ◦: partially or in a simplified form; –: not addressed).

Stream (representative works)	Dynamic human state	Rest as action	Partial observ.	Adaptive policy	Deadlines/ queue	Tuned cog. baselines
Dispatching rules [1, 2, 5, 72]	–	–	–	–	✓	–
Fatigue/recovery models in DRC systems [30–32]	✓	◦	–	–	◦	–
Rest-break / rest-allowance optimisation [33–35]	◦	✓	–	–	–	–
DRL for machine scheduling [40, 42–44, 46]	–	–	–	✓	✓	–
Fatigue-aware shop schedul- ing [50, 51]	✓	–	–	◦	✓	–
Adaptive automation / inter- ruption mgmt. [107, 112– 114]	✓	◦	✓	◦	–	–
This study	✓	✓	✓	✓	✓	✓

191 the remaining six arrive at slots drawn uniformly from $\{1, \dots, \lfloor 0.6T \rfloor\}$, which
 192 models interruptions and new requests. Task i has difficulty $d_i \sim \mathcal{U}[0.2, 1.0]$ and
 193 is labelled *hard* if $d_i \geq 0.6$ and *easy* otherwise. Its work content is $w_i = 1 + 2.5 d_i$
 194 work units, so a task requires between 1.5 and 3.5 slots (15–35 min) at full capacity.
 195 Its deadline is $D_i = \min\{T, a_i + \lceil s_i T \rceil\}$ with relative slack $s_i \sim \mathcal{U}[0.3, 1.0]$. With
 196 these settings, the expected daily work content (≈ 30 units) is below the capacity
 197 of a fully rested worker (48 units) but above what a worker can deliver once

Table 2: Notation.

Symbol	Meaning
$t \in \{0, \dots, T-1\}, \Delta$	decision slot index and slot length ($T = 48, \Delta = 10$ min)
h_t	clock time of slot t (h), $h_t = 9 + t\Delta/60$
a_i, d_i, w_i, D_i	arrival slot, difficulty, work content and deadline of task i
H_t	homeostatic attention resource (latent)
$c(h), A_t$	circadian modulation and effective attention, $A_t = \text{clip}(H_t + \gamma c(h_t))$
F_t	fatigue (latent)
ρ_t	work rate (work units completed per slot)
$\tilde{A}_t, \tilde{F}_t$	noisy observations of A_t, F_t
o_t, x_t	observation vector and history window of length k
u_t, b_t	indicators of working and resting in slot t
r_t	scalar reward; λ, μ fatigue and attention weights
γ_{RL}	RL discount factor (distinct from circadian amplitude γ)

198 attention and fatigue degrade. Scheduling decisions therefore matter: the day is
 199 *conditionally overloaded*.

200 3.2. Cognitive dynamics

201 *Effective attention.* Following the decomposition of alertness into homeostatic and
 202 circadian components [14, 96, 98], effective attention is

$$A_t = \text{clip}(H_t + \gamma c(h_t), 0, 1), \quad c(h) = 0.2 e^{-(h-10.5)^2/(2 \cdot 1.5^2)} - e^{-(h-14)^2/(2 \cdot 1.2^2)}, \quad (1)$$

203 where $c(h)$ encodes a late-morning peak and the post-lunch dip around 14:00
 204 [15–17] (Fig. 1a), and $\gamma = 0.15$ sets its amplitude.

205 *Homeostatic attention resource.* Working on a task of difficulty d depletes the
 206 attentional resource, and the depletion is accelerated by fatigue. This captures the
 207 vigilance decrement and resource-control accounts of sustained attention [6–8].
 208 Resting restores the resource exponentially towards its maximum [22, 93]:

$$H_{t+1} = \text{clip}\left(H_t - \alpha d (1 + \phi F_t) u_t + \beta (1 - H_t) b_t + \sigma_p \xi_t, 0, 1\right), \quad \xi_t \sim \mathcal{N}(0, 1), \quad (2)$$

209 where $u_t, b_t \in \{0, 1\}$ indicate work and rest ($u_t + b_t = 1$), $\alpha = 0.02$ is the depletion
 210 rate per unit difficulty, $\phi = 0.5$ the fatigue coupling, $\beta = 0.30$ the per-slot recovery
 211 fraction, and $\sigma_p = 0.02$ the process-noise scale that represents unmodelled day-
 212 to-day and moment-to-moment variability.

213 *Fatigue.* Fatigue accumulates exponentially towards saturation during work and
 214 decays exponentially during rest. This is the discrete-time analogue of the fatigue-
 215 recovery model of Jaber and Neumann [30], also used in [31, 32]:

$$F_{t+1} = \text{clip}\left(F_t + \delta d (1 - F_t) u_t - \eta F_t b_t, 0, 1\right), \quad (3)$$

216 with accumulation rate $\delta = 0.05$ and recovery rate $\eta = 0.20$.

217 *Productivity.* While working on task i , the worker completes

$$\rho_t = \rho_0 A_t^{0.5+d_i} (1 - \kappa F_t) \quad (4)$$

218 work units in slot t , with $\rho_0 = 1$ and $\kappa = 0.5$. The exponent makes hard tasks more
 219 sensitive to attention than easy ones (Fig. S1 in the Supplementary Material), in
 220 line with resource theories in which demanding tasks draw more heavily on limited
 221 capacity [90, 91]. The fatigue factor reduces throughput multiplicatively [10, 12].
 222 A task is complete at the end of the first slot in which its cumulative work reaches
 223 w_i .

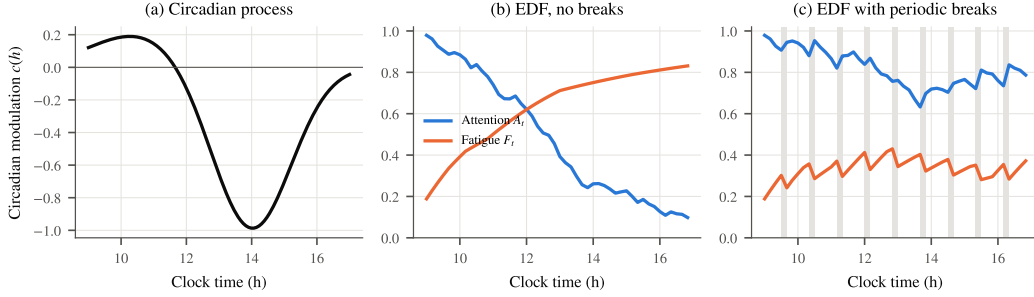


Figure 1: Simulator dynamics. (a) Circadian modulation $c(h)$ with a late-morning peak and post-lunch dip. (b) Effective attention and fatigue on one held-out test day under EDF without breaks. (c) The same day under EDF with a break after every four work slots (shaded bands mark breaks). Breaks keep attention high and fatigue low at the cost of working time.

224 *Initial state and behavioural plausibility.* Each day starts with $H_0 \sim \mathcal{U}[0.8, 1.0]$
 225 and $F_0 \sim \mathcal{U}[0, 0.15]$. Parameters were chosen *a priori* to reproduce qualitative
 226 patterns reported in the time-on-task and fatigue literature rather than fitted to a
 227 particular dataset. Under continuous work without breaks, attention falls from
 228 ≈ 1.0 to ≈ 0.3 by the end of the day, with a visible early-afternoon trough, and
 229 fatigue rises to ≈ 0.85 . One 10-minute break recovers 30% of the attentional deficit
 230 and 20% of accumulated fatigue (Fig. 1b,c). Because no single parameterisation
 231 can represent all workers, Section ?? evaluates the policies zero-shot on resilient
 232 and fatigue-prone profiles in which all four rate parameters are scaled by $\pm 30\%$.

233 3.3. Observation model and partial observability

234 The scheduler cannot observe H_t , A_t or F_t . It receives noisy estimates

$$\tilde{A}_t = \text{clip}(A_t + \sigma_o \epsilon_t^A, 0, 1), \quad \tilde{F}_t = \text{clip}(F_t + \sigma_o \epsilon_t^F, 0, 1), \quad \epsilon_t^A, \epsilon_t^F \sim \mathcal{N}(0, 1), \quad (5)$$

235 with $\sigma_o = 0.10$ in the nominal setting. These estimates stand in for wearable, oc-
 236 ular, behavioural or self-report estimators, whose errors are substantial in practice

237 [52–54]. The full observation is the 11-dimensional vector

$$o_t = \left[\frac{t}{T}, \tilde{A}_t, \tilde{F}_t, \underbrace{\frac{n_t^e}{N}, \frac{W_t^e}{20}, \sigma_t^e, \frac{m_t^e}{N}}_{\text{easy class}}, \underbrace{\frac{n_t^h}{N}, \frac{W_t^h}{20}, \sigma_t^h, \frac{m_t^h}{N}}_{\text{hard class}} \right], \quad (6)$$

238 where, for each class, n_t is the number of available unfinished tasks, W_t their
 239 remaining work, $\sigma_t = \text{clip}((\min_i D_i - t)/T, -1, 1)$ the normalised slack of the
 240 most urgent task (1 if the class is empty), and m_t the number of overdue tasks.
 241 Unlike a binary completion vector, this summary is invariant to the number of tasks
 242 and exposes the deadline information that urgency-aware scheduling requires. The
 243 worker’s true state and the future arrivals are hidden, so the decision problem is
 244 a POMDP $\langle \mathcal{S}, \mathcal{A}, P, R, \Omega, O, \gamma_{\text{RL}} \rangle$ [59]. Here $\mathcal{S}$ contains the full task set and
 245 (H_t, F_t, t) , Ω is the observation space of Eq. (6), O is given by Eq. (5), and P by
 246 Eqs. (1)–(4) together with the arrival process.

247 3.4. Action space

248 The action set is $\mathcal{A} = \{\text{EASY}, \text{HARD}, \text{BREAK}\}$. Choosing **EASY** (**HARD**) dispatches
 249 the available easy (hard) task with the earliest deadline for the current slot. If the
 250 chosen class is empty, the earliest-deadline task of the other class is dispatched
 251 instead. **BREAK** allocates the slot to rest, and the worker also rests when no task
 252 is available. This abstraction is deliberate. It lets the agent decide the *intensity* of
 253 work relative to the worker’s state and when to rest, while delegating the choice of
 254 a specific task within a class to EDF, which is near-optimal for that sub-problem.
 255 It also keeps the action space independent of N . Tasks are preemptible at slot
 256 boundaries, which is common in knowledge work.

257 3.5. Reward

258 The per-slot reward combines task outcomes with cognitive terms:

$$r_t = \underbrace{\sum_{i \in C_t} (1 + d_i) [\mathbb{I}(t+1 \leq D_i) + 0.25 \mathbb{I}(t+1 > D_i)] - 0.5 |\mathcal{M}_t|}_{r_t^{\text{task}}} + \mu A_{t+1} - \lambda F_{t+1}, \quad (7)$$

259 where C_t is the set of tasks completed in slot t , $\mathcal{M}_t$ is the set of tasks whose deadline
 260 expires unfinished at the end of slot t , $\mu = 0.05$ and $\lambda = 0.10$. Harder tasks are
 261 worth more, late completion retains a quarter of the value, and each missed deadline
 262 costs 0.5. The attention and fatigue terms express the organisation’s preference for
 263 sustainable work. Because such terms can change the optimal policy [132], we (i)
 264 report reward-independent operational metrics (on-time rate, completion, fatigue
 265 exposure) alongside the return, (ii) train an agent with $\lambda = \mu = 0$ to show that
 266 the learned behaviour is driven by the dynamics rather than by the reward terms,
 267 and (iii) sweep λ to trace the productivity–fatigue Pareto front. The objective
 268 is to find a policy π that maps observation histories to actions and maximises
 269 $\mathbb{E}_\pi \left[\sum_{t=0}^{T-1} \gamma_{\text{RL}}^t r_t \right]$.

270 4. Attention-aware deep Q-network

271 4.1. From Q-learning to deep Q-networks

272 For a fully observable MDP, the optimal action-value function satisfies the Bell-
 273 man optimality equation $Q^*(s, a) = \mathbb{E}[r + \gamma_{\text{RL}} \max_{a'} Q^*(s', a') \mid s, a]$ [137, 138],
 274 and tabular Q-learning converges to Q^* under standard conditions [84]. With con-
 275 tinuous cognitive variables, a tabular representation is infeasible, so Q is approx-
 276 imated by a neural network $Q(\cdot, \cdot; \theta)$ trained to minimise the temporal-difference

277 (TD) error. The DQN of Mnih et al. [85] stabilises this nonlinear approximation
 278 with an experience replay buffer [128], which breaks the temporal correlation
 279 of samples, and a periodically synchronised target network $\bar{\theta}$. Convergence is
 280 not guaranteed for nonlinear function approximation [130], so we assess stability
 281 empirically across independent seeds.

282 4.2. Handling partial observability with an observation history

283 In a POMDP, the optimal policy is a function of the belief $b_t(s) = \Pr(s_t =$
 284 $s \mid o_{0:t}, a_{0:t-1})$ [57, 59]. Exact belief tracking is intractable for our continuous,
 285 nonlinear dynamics, and a memoryless policy $\pi(o_t)$ can be arbitrarily suboptimal
 286 [64]. Two structural properties of the scheduling problem make a short history
 287 informative. First, the latent state is persistent. H_t and F_t change slowly relative
 288 to the observation noise, so averaging several consecutive estimates reduces the
 289 error: for k independent observations of a quasi-static quantity, the standard error
 290 falls by a factor $\sqrt{k}$. Second, the trajectory of the observations, for example fatigue
 291 rising after consecutive hard tasks or attention recovering after a break, reveals the
 292 direction and rate of change, which a single noisy reading cannot. We therefore
 293 feed the network the window

$$x_t = [o_{t-k+1}, \dots, o_t] \in \mathbb{R}^{11k}, \quad k = 4, \quad (8)$$

294 zero-padded at the start of the day. This is the finite-memory approximation used
 295 for Atari [85]. It is computationally cheaper than recurrent alternatives [65, 127]
 296 and is a strong baseline for many POMDPs [66]. A 40-minute window spans the
 297 time scale of a typical work bout. Section ?? varies $k \in \{1, 2, 4, 8\}$.

298 4.3. Double Q -learning and dueling architecture

299 Max-based targets overestimate action values in noisy environments [67], and
 300 our environment is noisy by design. AA-DQN therefore uses the Double-DQN
 301 target, which decouples action selection (online network) from evaluation (target
 302 network):

$$y_t = r_t + \gamma_{\text{RL}}(1 - \text{done}_t) Q(x_{t+1}, \arg \max_{a'} Q(x_{t+1}, a'; \theta); \bar{\theta}). \quad (9)$$

303 In many states, for example at the start of the day or when the queue is empty, the
 304 choice between easy and hard work matters much less than the value of the state
 305 itself. A dueling head [68] represents this explicitly:

$$Q(x, a; \theta) = V(x; \theta) + \left(A(x, a; \theta) - \frac{1}{|\mathcal{A}|} \sum_{a'} A(x, a'; \theta) \right). \quad (10)$$

306 The shared body consists of two fully connected layers of 128 ReLU units. The
 307 loss is the Huber loss of the TD error, which is robust to the occasional large
 308 reward of a hard task completed on time [139]:

$$\mathcal{L}(\theta) = \mathbb{E}_{(x,a,r,x') \sim \mathcal{D}} [\ell_{\delta=1}(Q(x, a; \theta) - y)]. \quad (11)$$

309 Parameters are updated with Adam [140] and gradient-norm clipping. The vanilla
 310 DQN baseline differs from AA-DQN only in using $k = 1$, the standard max target
 311 and a single linear output head. This isolates the contribution of memory and of
 312 the two stabilisation components.

313 4.4. Exploration and training procedure

314 Actions are selected ϵ -greedily with an exponentially decaying exploration
 315 rate $\epsilon_n = \epsilon_{\min} + (\epsilon_{\max} - \epsilon_{\min})e^{-\kappa_{\epsilon} n}$, where n is the environment step, $\epsilon_{\max} = 1$
 316 and $\epsilon_{\min} = 0.01$, and κ_{ϵ} is set so that ϵ reaches 0.05 after half of the training

Algorithm 1 Training of the attention-aware DQN (AA-DQN).

Require: episodes E , history length k , replay capacity $|\mathcal{D}|$, batch size B , target period C , validation days $\mathcal{V}$

- 1: initialise θ randomly, $\bar{\theta} \leftarrow \theta$, $\mathcal{D} \leftarrow \emptyset$, $n \leftarrow 0$, $J^* \leftarrow -\infty$
- 2: **for** episode $e = 1, \dots, E$ **do**
- 3: sample a workday (tasks, initial state, noise); observe o_0 ; $x_0 \leftarrow [\mathbf{0}, \dots, \mathbf{0}, o_0]$
- 4: **for** $t = 0, \dots, T - 1$ **do**
- 5: $a_t \leftarrow$ random action with prob. ϵ_n , else $\arg \max_a Q(x_t, a; \theta)$
- 6: dispatch a_t (EDF within class or rest); simulate Eqs. (1)–(4); receive r_t (Eq. (7))
and o_{t+1}
- 7: $x_{t+1} \leftarrow \text{shift}(x_t, o_{t+1})$; store $(x_t, a_t, r_t, x_{t+1}, done_t)$ in $\mathcal{D}$; $n \leftarrow n + 1$
- 8: **if** $|\mathcal{D}| \geq \text{warm-up}$ **then**
- 9: sample a minibatch of B transitions; compute targets with Eq. (9)
- 10: $\theta \leftarrow \theta - \eta_{lr} \text{Adam}(\nabla_{\theta} \mathcal{L}(\theta))$ with gradient clipping
- 11: **if** $n \bmod C = 0$ **then** $\bar{\theta} \leftarrow \theta$
- 12: **if** $e \bmod 100 = 0$ **then**
- 13: $J \leftarrow$ mean greedy return on $\mathcal{V}$; **if** $J > J^*$ **then** $J^* \leftarrow J$, $\theta^* \leftarrow \theta$
- 14: **return** θ^*

317 budget. Algorithm 1 summarises training. Every 100 episodes, the greedy policy
318 is evaluated on a fixed set of 100 *validation* days that are disjoint from the training
319 and test days, and the parameters with the highest validation return are retained.
320 This mirrors the tuning of the heuristic baselines on their own held-out days
321 (Section 5.1), so that both families receive the same model-selection budget and
322 neither sees the test days. For transparency we also report the final-iterate policy
323 (Supplementary Table S3).

324 4.5. Computational complexity

325 A forward pass costs $O(11k \cdot 128 + 128^2 + 128 \cdot |\mathcal{A}|) \approx 2.3 \times 10^4$ multiply–
326 accumulate operations, independent of the number of tasks, because the observa–
327 tion summarises the queue per class. The dispatch step is $O(N \log N)$. Inference
328 therefore takes microseconds on a CPU (Section ??), which is negligible relative to
329 a 10-minute decision interval and allows deployment on wearable or edge devices.

330 5. Experimental design

331 5.1. Baselines

332 We compare AA-DQN against the vanilla DQN and seven rule-based sched–
333 ulers, which span classical dispatching and hand-crafted cognition-aware policies.
334 Baselines dispatch individual tasks directly, so they have at least as much control
335 over task choice as the RL agents.

- 336 • **Random:** uniform choice among EASY, HARD and BREAK.
- 337 • **FIFO:** the available task that arrived first; never rests.
- 338 • **EDF:** the available task with the earliest deadline; never rests [2, 5].
- 339 • **SPT:** the available task with the least remaining work; never rests [1].
- 340 • **EDF-Periodic:** EDF with one break after every p work slots, a fixed work–
341 rest cycle in the spirit of ergonomic rest-allowance guidance [34].
- 342 • **EDF-Threshold:** EDF that rests whenever the *observed* fatigue exceeds τ_F
343 or the observed attention falls below τ_A . This is a reactive, state-aware rule.

344 • **Cog-Heuristic:** rests on the same threshold rule and otherwise schedules a
 345 hard task only when observed attention is at least τ_H , an easy task otherwise
 346 (EDF within class). This hand-crafted rule encodes the intuition that the
 347 learned policy might discover.

348 The parameters of the last three heuristics were selected by exhaustive grid
 349 search, maximising mean return on 300 *tuning* days that are disjoint from the
 350 training, validation and test days. The grids were $p \in \{2, 3, 4, 5, 6, 8, 10\}$;
 351 $\tau_F \in \{0.3, \dots, 0.9, 1.01\}$ and $\tau_A \in \{0, 0.2, \dots, 0.7\}$ (56 combinations); and,
 352 for Cog-Heuristic, $\tau_F \times \tau_A \times \tau_H$ with $\tau_H \in \{0.4, \dots, 0.9\}$ (252 combinations).
 353 The heuristics observe exactly the same noisy signals as the agents. The selected
 354 parameters are reported in Supplementary Table S2. Tuning is repeated separately
 355 for every environment variant, so that each ablation compares the agent with the
 356 best heuristic *for that variant*.

357 5.2. Protocol

358 Each RL configuration was trained with ten independent seeds (five for the
 359 reward-weight sweep and the component ablation, three for hyperparameter sen-
 360 sitivity) for 3000 episodes (1.44×10^5 decisions). Four disjoint sets of days were
 361 used: training days (generated from the training seed), 100 validation days for
 362 checkpoint selection, 300 tuning days for heuristic parameters, and 500 test days
 363 used only once, for the final evaluation. All methods are evaluated greedily on the
 364 *same* 500 test days with *common random numbers*. Each test day fixes the task set,
 365 the initial cognitive state and the sequences of process and observation noise, so
 366 differences between methods are paired and not confounded by sampling variation.
 367 In total, 102 agents were trained and $\sim 1.5 \times 10^5$ test days were simulated.

368 5.3. Metrics

369 Besides the return (Eq. (7)), which depends on the reward weights, we re-
370 port reward-independent operational metrics: the *on-time rate* (fraction of the
371 day’s tasks completed by their deadline); the *difficulty-weighted on-time rate*
372 $\sum_i d_i \mathbb{I}(\text{on time}) / \sum_i d_i$; the *completion rate* (completed at any time); the mean
373 *true* attention $\bar{A}$ and fatigue $\bar{F}$ over the day; the *high-fatigue exposure*, i.e. the
374 fraction of slots with $F_t > 0.7$; and the *break share*.

375 5.4. Statistical analysis

376 Following recommendations for reliable RL evaluation [60, 61], we treat the
377 training seed as the unit of replication for learned policies. For each seed, metrics
378 are averaged over the 500 test days. We report the mean and standard deviation
379 across seeds, and 95% bootstrap confidence intervals [141] with 5000 resamples.
380 Deterministic heuristics are summarised across test days. To compare AA-DQN
381 with each baseline, we average AA-DQN over seeds for each test day and apply the
382 two-sided Wilcoxon signed-rank test [142] to the 500 paired day-level differences.
383 We report the matched-pairs rank-biserial correlation r_{rb} as effect size, the win
384 rate, and Holm-adjusted p -values [143] over the seven baseline comparisons. As
385 a more conservative seed-level check, we also report how many of the ten seeds
386 outperform each baseline.

387 5.5. Implementation

388 The simulator, agents and baselines are implemented in Python with NumPy
389 and PyTorch [144]. Table 4 lists the hyperparameters. Training used one CPU
390 thread per run. The code, configuration files, raw per-day results and analysis

391 scripts are released (see Data availability) so that every number and figure in this
392 paper can be regenerated with a single command.

393 **6. Results**

394 *6.1. Learning dynamics*

395 Figure 2 shows greedy validation performance during training, averaged over
396 ten seeds. AA-DQN reaches the return of the best tuned heuristic after roughly
397 600 episodes, and the seed-averaged curve then fluctuates around that level. Its
398 on-time rate, however, rises above the heuristic within 400 episodes and stays
399 above it throughout. The seed average hides substantial checkpoint-to-checkpoint
400 variation within each run. Individual checkpoints exceed the heuristic return by a
401 clear margin, while others fall below it. This is the well-known policy oscillation
402 of value-based deep RL under a constant learning rate [60], and it motivates
403 selecting checkpoints on held-out validation days. Because validation and test
404 days are disjoint, the test results in Section 6.2 are not inflated by this selection.
405 The vanilla DQN learns equally fast at first but stalls below the heuristic. Its
406 *training* return even declines after about 500 episodes (Supplementary Fig. S2),
407 as the exploration rate decreases and the memoryless agent increasingly commits
408 to actions based on single noisy readings. This pattern is consistent with the over-
409 estimation that Double Q-learning is designed to reduce [67] and with the known
410 weakness of memoryless policies in POMDPs [64]. Validation-based selection
411 picked checkpoints at episode 1480 on average for AA-DQN and 950 for DQN. The
412 TD loss falls rapidly, rises moderately as value estimates grow, and then decreases
413 slowly (Fig. S2), the usual signature of bootstrapped value learning with a moving
414 target.

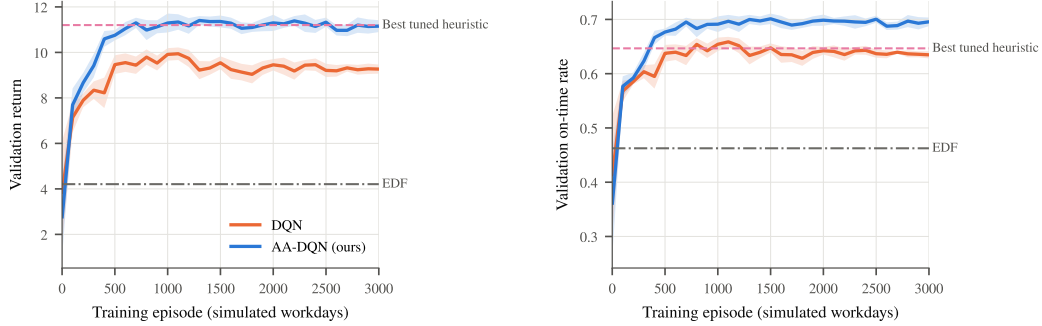


Figure 2: Greedy validation performance during training (mean of ten seeds; shaded bands are 95% confidence intervals). Horizontal lines show the best tuned heuristic (EDF-Periodic) and EDF on the same 100 validation days.

6.2. Comparison with baselines

Table 5 and Fig. 3 report performance on the 500 held-out test days. Three groups emerge. The rules that never rest (FIFO, EDF, SPT) drive the worker into the low-capacity regime: mean fatigue exceeds 0.47, attention falls to about 0.46–0.54, and EDF completes only 46.8% of tasks on time. EDF performs particularly poorly, which is the overload domino effect known from real-time scheduling [70]: once capacity falls, chasing the most urgent task means working on tasks that will be late anyway. SPT does better (62.5%) because finishing short tasks first limits the damage, but it still ignores the worker’s state. The three cognition-aware heuristics, which rest about 19–20% of the time, raise the on-time rate to 64–66% and reduce mean fatigue by almost half. This confirms that the dynamics reward rest, as intended.

AA-DQN achieves the highest return (12.19 ± 0.27) and the highest on-time rate (72.6%) of all methods. Relative to the best tuned heuristic (EDF-Periodic), the paired difference is +0.77 return (95% CI [0.58, 0.95]) and +6.6 percentage points of on-time completion ([5.7, 7.5]). Both are significant after Holm correction

431 ($p < 10^{-4}$; Table 6). All ten seeds individually outperform every baseline. The
 432 effect sizes are medium for return ($r_{rb} = 0.34$) and large for the on-time rate
 433 ($r_{rb} = 0.67$). The day-to-day variability of all methods is large (standard deviation
 434 of about 3 return units across days), because some days are intrinsically harder
 435 than others. AA-DQN therefore does not win every day. It obtains a strictly
 436 higher return than EDF-Periodic on 58.6% of days and a strictly higher on-time
 437 rate on 67.0%, with ties on most of the remaining days. Relative to the vanilla
 438 DQN, adding memory and the two stabilisation components increases return by
 439 1.53 ([1.46, 1.60]) and wins on 98.4% of days.

440 Two qualifications place the gain in context. First, AA-DQN rests *less* than the
 441 tuned heuristics (15.9% versus 18.8% of slots). It therefore operates at a somewhat
 442 higher mean fatigue (0.326 versus 0.296). This level is still 40% below EDF, and
 443 fatigue almost never exceeds 0.7 (high-fatigue exposure 0.0%, versus 30.4% under
 444 EDF). Second, AA-DQN completes *more tasks* on time while processing slightly
 445 *less* total work than EDF-Periodic. Its difficulty-weighted on-time rate (65.7%)
 446 equals that of EDF-Periodic (65.7%). The improvement thus comes from *which*
 447 tasks are worked on *when*, not from extracting more effort from the worker.
 448 Section ?? analyses this behaviour.

449 Validation-based checkpoint selection matters for the size of the margin over the
 450 heuristics (Supplementary Table S3). The final-iterate AA-DQN still completes
 451 more tasks on time than EDF-Periodic (70.6% versus 66.0%) but its return (11.35)
 452 is on a par with the heuristic. The final-iterate vanilla DQN (9.47) falls further
 453 behind. We report the validation-selected policy as the primary result because the
 454 heuristics received an equivalent selection step on their own held-out days.

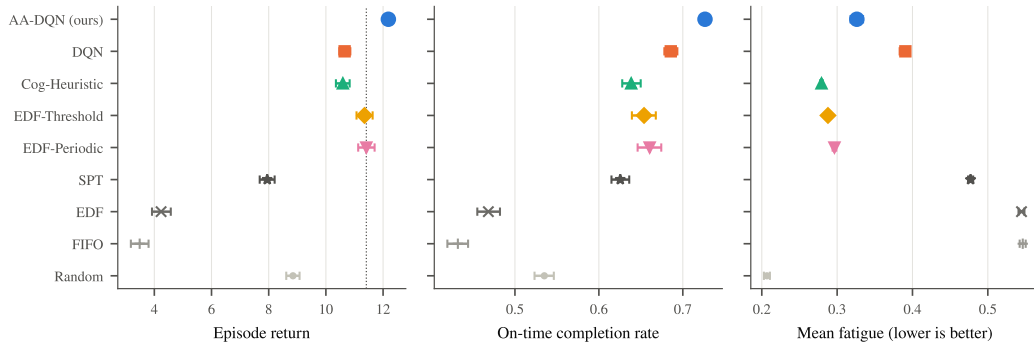


Figure 3: Test performance with 95% bootstrap confidence intervals (learned policies: across seeds; heuristics: across days). The dotted line marks the best heuristic return.

7. Discussion

7.1. Ethical and organisational considerations

A scheduler that monitors attention and fatigue and recommends when a person should work on what and when to rest touches on autonomy, privacy and fairness, and these issues must be designed in rather than added later [108, 112, 133]. We see four requirements for responsible deployment. First, *the system should be advisory*. Recommendations should be presented to the worker, who retains the authority to accept or override them; evidence from human–robot teaming indicates that the allocation of decision authority affects both efficiency and satisfaction [121]. Second, *cognitive-state data are sensitive personal data*. Physiological and behavioural signals should be processed with informed consent, ideally on the worker’s own device, and should never be used for performance appraisal. The observation model of AA-DQN needs only two scalar estimates per slot, which supports such data minimisation. Third, *fairness across workers* requires that policies do not systematically disadvantage people whose fatigue dynamics differ from the nominal profile. Section ?? shows how performance changes for resilient

471 and fatigue-prone profiles, and personalisation (Section 7.3) would further reduce
472 such disparities. Fourth, *safety constraints* should bound the policy, for example by
473 guaranteeing a minimum amount of rest per period regardless of deadline pressure,
474 consistent with safe RL principles [134]. Notably, the learned policies in this
475 study already increase rest relative to deadline-driven rules. The reward weight λ
476 makes the productivity–well-being trade-off explicit and auditable (Section ??), so
477 stakeholders can choose it deliberately rather than leaving it implicit in a heuristic.

478 7.2. Limitations

479 The study has limitations that define its scope. (i) *Simulation, not human data*.
480 The cognitive model is grounded in established functional forms and its parameters
481 were chosen to be behaviourally plausible, but it has not been calibrated against
482 individual time-series of attention and fatigue. The absolute numbers should
483 therefore be interpreted comparatively. The robustness analyses mitigate, but do
484 not remove, this concern. (ii) *Abstracted sensing*. Cognitive-state estimation is
485 modelled as additive Gaussian noise on the true state. Real estimators can be
486 biased, drift over the day, or fail intermittently, and they may be correlated with
487 task type. (iii) *A single worker and a single day*. Multi-day carry-over of fatigue,
488 sleep, and team-level interactions such as hand-offs and shared deadlines are
489 outside the scope. (iv) *Action abstraction*. The agent chooses a difficulty class and
490 delegates the choice of task within a class to EDF. This keeps the policy compact
491 and size-invariant but excludes policies that, for example, deliberately abandon a
492 hopeless task. (v) *Algorithmic scope*. We study value-based agents with a finite
493 history. Recurrent, model-based or policy-gradient agents [65, 127, 145] might
494 extract more from the observation sequence. (vi) *Simulated time-on-task effects*
495 *only*. Motivation, task interest, interruptions and errors, which also shape real

496 productivity [115, 116, 146], are not modelled.

497 7.3. Future work

498 Three extensions follow directly. First, *empirical calibration and validation*:
499 fitting the latent dynamics to longitudinal data from vigilance probes, wearables
500 and task logs [52, 55, 104], then evaluating the scheduler in a controlled user study
501 with NASA-TLX and performance outcomes [56], ideally as a micro-randomised
502 trial [117]. Second, *personalisation*: domain randomisation over worker profiles
503 during training [147, 148], and Bayesian or meta-learning adaptation to an individ-
504 ual’s parameters, in the spirit of personalised just-in-time interventions [118, 119].
505 Third, *multi-worker and human–robot settings*: extending the formulation to teams
506 and to collaborative cells in which a robot absorbs demanding sub-tasks when a
507 co-worker’s attention is low [122–125], with constrained or multi-objective RL to
508 enforce explicit well-being guarantees [63].

509 8. Conclusion

510 TBD

511 CRediT authorship contribution statement

512 **P.N. Bhargav Teja:** Conceptualization, Methodology, Software, Formal
513 analysis, Investigation, Visualization, Writing – original draft. **Lanka Sree**
514 **Chathurya:** Methodology, Software, Validation, Data curation, Investigation,
515 Writing – original draft. **Rajay Vedaraj I.S.:** Conceptualization, Supervision,
516 Methodology, Resources, Project administration, Writing – review & editing.

517 **Declaration of competing interest**

518 The authors declare that they have no known competing financial interests or
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524 **Data availability**

525 The simulator, agents, baselines, experiment configurations, raw per-episode
526 results and the scripts that generate every table and figure are openly available at
527 <https://github.com/bhargavteja-9779/r1>. No human-subject data were
528 collected.

529 **Declaration of generative AI and AI-assisted technologies in the manuscript** 530 **preparation process**

531 During the preparation of this work the authors used an AI assistant (Claude,
532 Anthropic) to support code development, language editing and the organisation of
533 the manuscript. After using this tool, the authors reviewed and edited the content
534 as needed and take full responsibility for the content of the published article.

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Table 3: Nominal simulator parameters and their rationale.

Parameter	Value	Rationale / source
Slots per day T , slot length Δ	48, 10 min	8-h workday; decision granularity of micro-breaks [22]
Tasks per day N (initial / arriving)	12 (6 / 6)	conditionally overloaded day; stochastic arrivals
Difficulty d_i , hard threshold	$\mathcal{U}[0.2, 1]$, 0.6	range of the original formulation
Work content w_i	$1 + 2.5d_i$ units	15–35 min at full capacity
Relative slack s_i	$\mathcal{U}[0.3, 1]$	range of the original formulation
Attention depletion α , fatigue coupling ϕ	0.02, 0.5	vigilance decrement over hours of work [7, 8]
Attention recovery β	0.30 per slot	break benefits [22, 93]
Circadian amplitude γ	0.15	post-lunch dip [15, 16]
Fatigue accumulation δ , recovery η	0.05, 0.20	exponential fatigue–recovery [30, 31]
Fatigue throughput penalty κ	0.5	performance loss under fatigue [10, 136]
Process / observation noise σ_p, σ_o	0.02, 0.10	variability; sensor error [52, 54]
Reward weights μ, λ	0.05, 0.10	swept in Section ??

Table 4: Agent hyperparameters (shared by DQN and AA-DQN unless stated).

Hyperparameter	DQN	AA-DQN
History length k	1	4
Target	max (standard)	Double DQN
Output head	linear	dueling ($V + \text{advantage}$)
Hidden layers		2×128 , ReLU
Optimiser, learning rate		Adam, 5×10^{-4}
Discount γ_{RL}		0.99
Replay capacity, warm-up		50 000, 1 000 transitions
Minibatch size		64
Target update period C		1 000 steps (hard copy)
Loss, gradient clip		Huber ($\delta = 1$), $\ \nabla\ _2 \leq 10$
Exploration	ϵ : $1.0 \rightarrow 0.01$, exponential (0.05 at 50% of budget)	
Training budget	3 000 episodes = 144 000 decisions	
Checkpoint selection	best greedy return on 100 validation days, every 100 episodes	

Table 5: Performance on 500 held-out test days (nominal environment). Learned policies: mean $\pm$ s.d. across ten training seeds of the per-seed test means. Heuristics: mean $\pm$ s.d. across test days (they are deterministic given the day). Bold marks the best value.

Method	Return	On-time rate	Weighted on-time	Mean attention	Mean fatigue	Break share
Random	8.85 $\pm$ 2.71	0.535 $\pm$ 0.131	0.488 $\pm$ 0.143	0.855 $\pm$ 0.037	0.207 $\pm$ 0.045	0.326 $\pm$ 0.065
FIFO	3.49 $\pm$ 3.49	0.432 $\pm$ 0.139	0.422 $\pm$ 0.125	0.463 $\pm$ 0.106	0.546 $\pm$ 0.045	0.000 $\pm$ 0.003
EDF	4.24 $\pm$ 3.82	0.468 $\pm$ 0.155	0.454 $\pm$ 0.137	0.465 $\pm$ 0.105	0.545 $\pm$ 0.046	0.000 $\pm$ 0.003
SPT	7.95 $\pm$ 2.94	0.625 $\pm$ 0.120	0.494 $\pm$ 0.122	0.543 $\pm$ 0.093	0.477 $\pm$ 0.038	0.000 $\pm$ 0.001
EDF-Periodic	11.41 $\pm$ 3.32	0.660 $\pm$ 0.160	0.657 $\pm$ 0.148	0.798 $\pm$ 0.040	0.296 $\pm$ 0.029	0.188 $\pm$ 0.005
EDF-Threshold	11.35 $\pm$ 3.28	0.654 $\pm$ 0.164	0.650 $\pm$ 0.155	0.806 $\pm$ 0.022	0.288 $\pm$ 0.034	0.201 $\pm$ 0.037
Cog-Heuristic	10.59 $\pm$ 2.76	0.638 $\pm$ 0.132	0.533 $\pm$ 0.150	0.803 $\pm$ 0.034	0.279 $\pm$ 0.020	0.198 $\pm$ 0.025
DQN	10.66 $\pm$ 0.33	0.686 $\pm$ 0.013	0.606 $\pm$ 0.016	0.692 $\pm$ 0.011	0.391 $\pm$ 0.011	0.105 $\pm$ 0.018
AA-DQN (ours)	12.19 $\pm$ 0.27	0.726 $\pm$ 0.009	0.657 $\pm$ 0.012	0.761 $\pm$ 0.013	0.326 $\pm$ 0.014	0.159 $\pm$ 0.015

Table 6: Paired comparison of AA-DQN with each baseline on 500 common-random-number test days: mean difference with 95% bootstrap CI, matched-pairs rank-biserial correlation r_{rb} , and Holm-adjusted Wilcoxon signed-rank p -values.

AA-DQN vs.	Episode return			On-time completion rate		
	Δ [95% CI]	r_{rb}	p_{Holm}	Δ [95% CI]	r_{rb}	p_{Holm}
Random	+3.34 [+3.13, +3.55]	0.97	$< 10^{-4}$	+0.191 [+0.182, +0.202]	1.00	$< 10^{-4}$
FIFO	+8.70 [+8.47, +8.92]	1.00	$< 10^{-4}$	+0.294 [+0.286, +0.303]	1.00	$< 10^{-4}$
EDF	+7.95 [+7.71, +8.18]	1.00	$< 10^{-4}$	+0.258 [+0.249, +0.267]	1.00	$< 10^{-4}$
SPT	+4.24 [+4.06, +4.41]	1.00	$< 10^{-4}$	+0.101 [+0.094, +0.108]	0.93	$< 10^{-4}$
EDF-Periodic	+0.77 [+0.58, +0.95]	0.34	$< 10^{-4}$	+0.066 [+0.057, +0.075]	0.67	$< 10^{-4}$
EDF-Threshold	+0.84 [+0.66, +1.02]	0.37	$< 10^{-4}$	+0.073 [+0.064, +0.082]	0.71	$< 10^{-4}$
Cog-Heuristic	+1.59 [+1.45, +1.74]	0.83	$< 10^{-4}$	+0.088 [+0.082, +0.094]	0.93	$< 10^{-4}$